KEY LEMMAS (for verification)

MO 413935 · $L = \lim \alpha_n = \frac{1}{2}$ · three load-bearing statements

SETUP

$n = p^2 + 1$, p odd prime; C Paley conference; $\Phi(C) = \frac{1}{2} n \sqrt{n-1}$.

Gap-2 undercutter: edge-set F with $\Phi(C \oplus F) = \Phi(C) - 2$.

Goal: no such F descends further $\Rightarrow m_n \geq \Phi(C) - 2 \Rightarrow L = \frac{1}{2}$

(denseness of $\{p^2+1\}$ + universal $\limsup \alpha_n \leq \frac{1}{2}$).

LEMMA A — bi-tight covers empty (Prop 15.167)

Let G be the Max± edge Gram, $d = n/2$, $\mathrm{mult}(\lambda_{\max}) \geq d - 1$, $\lambda_{\min} \geq 6$.

Majorization upper-bounds the top eigenvalue:

$$L_* = \frac{p^4 + 24p^2 - 1}{2(p^2-1)}.$$

Algebra for primes $p \geq 5$:

$$2d - L_* = \frac{p^4 - 24p^2 - 1}{2(p^2-1)} > 0 \quad \text{(numerator 24 at } p = 5\text{; increasing thereafter).}$$

Hence $\lambda_{\max} \leq L_* < 2d \;\Rightarrow\; \lambda_{\text{cycle}} < d \;\Rightarrow\; \lambda_{\max}(G) = n/2$ simple

$\Rightarrow$ no bi-tight size-$2p$ cover (Prop 15.55). Residual / $16N$ not used.

CHECK (algebra only)

- At $p = 5$: $L_* = (625 + 600 - 1)/(2 \cdot 24) = 1224/48 = 25.5$, $2d = 26$, gap $= 0.5$.
- Poly $f(x) = x^2 - 24x - 1 > 0$ for $x = p^2 \geq 25$.
- Chain: $\text{mult} + \lambda_{\min} \;\Rightarrow\; L_* \;\Rightarrow\; \lambda_{\text{cycle}} < d \;\Rightarrow\;$ bi-tight empty.

Key lemmas only — full writeup separate

LEMMA B — Type I freeness-fail (Prop 15.170)

Type I, $k = 3p - 2$, freeness fail, gap-2 $\Rightarrow$ $s_- = -1$. Bad case dualizes to

$$(\mathrm{Gsum}\, x)_e = 6/p - 4, \quad \text{with } 0 \le x_f \le 1,\ \sum x = k,\ x_e = 0.$$

Box–sum lower bound: $(\mathrm{Gsum}\, x)_e \ge -12k/(p\,n)$.

Need $6/p - 4 < -12k/(pn)$. Substitute $k = 3p - 2$, $n = p^2 + 1$:

$$4p^3 - 6p^2 - 32p + 18 > 0 \text{ for all primes } p \ge 5 \ \ (\text{check } p = 5\text{: } 500 - 150 - 160 + 18 = 208 > 0).$$

Contradiction $\Rightarrow$ freeness-fail Type I with $s_- \le -1$ empty.

LEMMA C — deep freeness-fail (Prop 15.171)

Deep class: $s_+ = 2$, freeness fail, $k \ge 3p$.

- Free covers: weak ND (no further descent).
- Auto-freeness for $k \le 3p - 2$; fail-equality at $k = 3p - 1$ empty.
- Remaining freeness-fail dualizes to two-level equality

$$(\mathrm{Gsum}\, x)_e = 2(8 - 3k/p), \ \text{same box LB } -12k/(pn).$$

Obstructed for every prime $p \ge 5$ $\Rightarrow$ deep freeness-fail ND.

ASSEMBLY

A + freeness / tight empty + B + C $\Rightarrow$ $m_n \ge \Phi(C) - 2$ on $n = p^2 + 1$.

Denseness (Prop 6.2) + $\limsup \alpha_n \le \frac{1}{2}$ $\Rightarrow$ $L = \frac{1}{2}$.

Not claimed here: residual / $16N$ spectral package (optional; not required for L).